

Kalivarapu Charan Sai

+91-7981099712 | kalivarapucharan48@gmail.com | linkedin.com/in/charan-kalivarapu-b64ab0349 | Hyderabad, India

SUMMARY

Motivated AI & ML undergraduate with hands-on experience designing and deploying end-to-end machine learning pipelines, computer vision systems, and NLP-driven applications. Proficient in Python, TensorFlow, Scikit-learn, and REST API integration. Eager to contribute to real-world AI/ML engineering projects by building scalable, data-driven intelligent systems.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML, CSS, C

ML / AI Frameworks & Libraries: TensorFlow, Scikit-learn, CNNs, Whisper STT, T5, GPT-J, REST APIs

MLOps & DevOps Tools: Docker, Git, GitHub, Streamlit, Google Colab, Jupyter Notebook, VS Code

Specializations: Deep Learning, Computer Vision, NLP, Prompt Engineering, End-to-End ML Pipelines

Other: Bootstrap, LaTeX, Data Preprocessing, Model Evaluation, Adaptive AI Systems

PROJECTS

- AI Diagnostic System — Medical Image Classification** | *Python, TensorFlow, CNN* | [GitHub](#) Jan 2026
- Designed and deployed an end-to-end deep learning pipeline for automated disease detection from medical images, covering preprocessing, model training, evaluation, and prediction interface.
 - Implemented CNN-based architectures for feature extraction and multi-class classification, achieving **92% classification accuracy** on the validation dataset.
 - Built a structured model evaluation framework benchmarking performance metrics (accuracy, precision, recall, F1-score), attaining **89% precision** and **91% recall** across test classes.
- AI-Driven Smart Traffic Signal System with Emergency Priority** | *Python, Computer Vision, IoT* Aug 2025
- Built a real-time AI traffic management system integrating computer vision and multi-sensor inputs (IR, acoustic, loop sensors) to dynamically optimize signal timings based on live traffic density (600–800 m range).
 - Implemented emergency vehicle detection with **95% detection accuracy** and directional priority control (N/S/E/W), autonomously clearing routes and reducing average congestion response time by **40%**.
 - Achieved **93% overall signal optimization accuracy** by fusing traffic camera feeds and multi-sensor data for robust real-time decision-making across varying traffic conditions.
- AI Training Assistant — Intelligent Learning Support System** | *Python, NLP, GPT-J, T5* | [GitHub](#) Feb 2025
- Developed an AI-powered personalized learning assistant leveraging NLP models (T5, GPT-J) for automated query handling and adaptive training content delivery, achieving **88% query response accuracy**.
 - Designed modules for intelligent guidance and training efficiency optimization using prompt engineering, improving content relevance scores by **35%** based on user feedback evaluation.
- OpenDoor — Smart Service & Access Management Platform** | *Python, REST APIs, Docker* | [GitHub](#) Aug - 2024
- Developed a scalable web-based platform with structured backend logic for secure user access control, request tracking, and service management workflows, supporting **99% uptime** during testing.
 - Integrated REST API endpoints with an average response time of **under 200ms** and containerized the application using Docker for reproducible and scalable deployment.
- WanderList — Travel Planning Web Application** | *JavaScript, HTML, CSS, Bootstrap* | [GitHub](#) Jan - 2024
- Built a responsive travel planning web app featuring destination listing, itinerary management, and structured data handling, achieving **95% mobile responsiveness score** across devices and reducing page load time by **30%** through optimized data handling.
- Crop Price Prediction System** | *Python, Scikit-learn, Pandas* | [GitHub](#) Aug - 2023
- Built a machine learning model to predict crop prices using historical agricultural data, enabling data-driven insights for farmers and stakeholders.
 - Applied regression algorithms and feature engineering techniques to achieve reliable price forecasting with optimized model performance.

EDUCATION

Woxsen University

B.Tech in Artificial Intelligence & Machine Learning

Narayana College

Intermediate — MPC

Hyderabad, India

2023 – Present — CGPA: 7.6

2021 – 2023 — 60%